

QBITEL BRIDGE

Banking & Financial Services

DEPLOYMENT CHECKLIST

Banking & Financial Services

10 Phases | 16 Weeks | ISO 8583 | SWIFT | COBOL | FIX | PCI-DSS 4.0 | DORA

WK 0-2

Discover
& Plan

WK 3-5

Build
Infra

WK 6-10

Protect
Payments

WK 11-16

Go
Live

0
PHASE

PRE-ENGAGEMENT & SCOPING

Week 0 (Pre-Start) | Owner: Account Executive + Solutions Architect

Phase 0 must be completed before any on-site deployment begins. All access approvals must be in writing.

☐ Confirm executive sponsor: CISO or CTO with authority to approve scope and budget.

CRITICAL

- Identify CISO, CTO, Head of Payments, Head of Compliance as primary stakeholders
- Confirm board or ExCo awareness of quantum security initiative
- Assign internal QBITEL Bridge project owner (minimum 0.5 FTE)

☐ Define engagement scope: which payment rails, protocols, and data centers are in scope for Phase 1.

REQUIRED

- List all ISO 8583 card scheme connections (Visa, Mastercard, domestic)
- List all SWIFT connectivity points (Alliance Access, Alliance Gateway)
- Identify mainframe environments (z/OS version, CICS/IMS, DB2/IMS DB)
- Identify FIX trading environments if applicable

☐ Conduct infrastructure audit: collect existing network diagrams, HSM inventory, and protocol documentation.

REQUIRED

- Request existing network diagrams for payment DMZ and mainframe LAN
- Collect HSM inventory: vendor, model, firmware version, partition structure
- Gather SWIFT CSP last attestation report
- Collect most recent PCI-DSS QSA report or SAQ

☐ Complete compliance gap assessment: PCI-DSS 4.0, DORA, SWIFT CSP 2025.

REQUIRED

- Map current cryptographic implementations against PCI-DSS 4.0 Requirements 3, 4, 12.3.3
- Assess DORA Article 9/10/11 current posture
- Review SWIFT CSP mandatory controls status

☐ Obtain data center access approvals for passive tap deployment.

REQUIRED

- Submit change request for SPAN/mirror port configuration on payment network switches
- Obtain network security team sign-off for passive monitoring

☐

Define success criteria and KPIs for the engagement.

REQUIRED

- Agree on TPS throughput target (default: 10,000 TPS)
- Agree on latency budget (default: <50ms)
- Agree on compliance evidence package format and target date

☐

Complete QBITEL Bridge master service agreement and data processing agreement.

REQUIRED

- Legal review of MSA, DPA, SLA schedules
- GDPR data processing agreement for any EU customer data in scope
- Escrow arrangement documentation (if required)

☐

Schedule kickoff workshop with all stakeholders.

REQUIRED

- 2-day on-site workshop: architecture review, protocol discovery approach, risk register seed

1

PHASE

PROTOCOL DISCOVERY

Weeks 1-2 | Owner: Solutions Architect + Client Network Team

Protocol discovery is passive - zero traffic modification. SPAN configuration is the only change required to production infrastructure.

☐

Configure SPAN/mirror ports on payment network core switches for passive tap.

REQUIRED

- Payment DMZ switch: configure SPAN for all ports carrying ISO 8583 traffic
- Mainframe LAN switch: configure SPAN for TN3270e and DRDA traffic
- SWIFT zone switch: configure SPAN for SWIFT Alliance Access connections
- Trading LAN switch: configure SPAN for FIX session ports (if in scope)

☐

Deploy QBITEL Bridge Discovery Appliance in passive-only mode.

REQUIRED

- Rack-mount or VM deployment in out-of-band monitoring VLAN
- Verify no traffic modification capability - discovery appliance is receive-only
- Configure packet capture storage: minimum 2TB for 48-hour capture baseline

☐

Execute automated ISO 8583 and ISO 20022 protocol fingerprinting.

REQUIRED

- Confirm ISO 8583 bit-map detection across all card scheme connections
- Identify ISO 20022 message types in use (pacs.008, camt.053, pain.001)
- Map acquirer-to-issuer and processor-to-scheme message flows
- Identify any legacy ISO 8583 v1987 vs v1993 vs v2003 versions in use

☐

Execute automated SWIFT MT/MX protocol fingerprinting.

REQUIRED

- Identify SWIFT MT103, MT202, MT515, MT940, MT950 message flows
- Confirm FIN framing vs MX ISO 20022 identification
- Map SWIFT Alliance Access connectivity paths
- Identify any proprietary SWIFT extensions in use

☐

Execute TN3270e and mainframe protocol discovery.

REQUIRED

- Identify TN3270e sessions: source IP, CICS region, transaction codes
- Identify DB2 DRDA connection sources and database names
- Identify IBM MQ queue managers and channels carrying sensitive data
- Map IMS Connect connection paths if applicable

☐

Execute FIX protocol discovery if trading in scope.

RECOMMENDED

- Identify FIX versions (4.2, 4.4, 5.0) and ComplIDs in use
- Map FIX sessions to ECN/exchange counterparties
- Identify FpML connections if OTC derivatives in scope

☐

Generate Protocol Risk Register.

REQUIRED

- Rank all discovered protocols by quantum vulnerability (RSA key size, cipher suite)
- Assign breach impact score (payment volume, customer PII, regulatory sensitivity)
- Calculate quantum vulnerability timeline per protocol
- Produce prioritized remediation roadmap

☐

Present Protocol Risk Register to CISO and Head of Payments.

REQUIRED

- Schedule executive briefing: 60-minute presentation of discovery findings
- Agree remediation priority order for Phase 3
- Obtain sign-off on Phase 2 scope

2
PHASE**INFRASTRUCTURE READINESS**

Weeks 3-5 | Owner: Solutions Architect + Client Infrastructure Team

Phase 2 introduces QBITEL Bridge into the network but in passive mode only. No traffic modification until Phase 3 activation.

☐

Provision HSM cluster (minimum 2 HSMs for HA).

REQUIRED

- Option A - Thales Luna Network HSM 7: rack, cable, initialize, partition for QBITEL Bridge
- Option B - AWS CloudHSM: provision cluster in payment VPC, configure PKCS#11 client
- Option C - Azure Managed HSM: provision in Azure payment subscription
- Generate root key material within HSM boundary - never exported

☐

Configure HSM network partitions and access controls.

REQUIRED

- Create dedicated QBITEL Bridge application partition
- Configure HSM admin credentials in HSM-backed vault (not plaintext)
- Enable FIPS 140-3 Level 3 operation mode
- Configure HSM audit logging to tamper-evident syslog

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Deploy QBITEL Bridge main appliances (active-active HA pair).

REQUIRED

- Rack-mount or VM deployment in payment DMZ
- Configure active-active clustering with synchronous state replication
- Verify PKCS#11 connectivity to HSM cluster
- Configure management interface with RBAC and MFA

☐

Configure network segmentation for PCI cardholder data environment.

REQUIRED

- Verify QBITEL Bridge placement between card network gateway and core banking
- Confirm no direct cardholder data flows bypass QBITEL Bridge
- Configure ACLs to restrict QBITEL Bridge management access
- Document network topology changes for PCI-DSS scoping

☐

Enable database audit logging for DB2 and IMS DB.

RECOMMENDED

- Enable DB2 audit policy for QBITEL Bridge service account connections
- Configure DB2 DRDA logging for connection authentication events
- Enable IMS Connect audit for remote connection events if applicable

☐ Configure SIEM integration (Splunk / QRadar / Sentinel).

REQUIRED

- Install QBITEL Bridge Splunk app or configure CEF/LEEF output
- Configure QBITEL Bridge event forwarding to SIEM HEC/syslog
- Validate PCI-DSS and DORA dashboard data flow
- Test alert delivery to SOC ticketing system

☐ Conduct integration testing in passive (read-only) mode.

REQUIRED

- Verify QBITEL Bridge receives all protocol flows from SPAN
- Verify HSM operations are functional (key generation, PKCS#11 test)
- Verify SIEM receives events from QBITEL Bridge
- Verify compliance dashboards populate correctly

☐ Staff training: operations team, SOC analysts, compliance team.

REQUIRED

- QBITEL Bridge operator training: 1-day hands-on session
- SOC analyst training: alert triage, playbook execution, escalation
- Compliance team training: evidence portal, report generation

3

PHASE

SECURITY POLICY CONFIGURATION

Weeks 4-5 (parallel with Phase 2) | Owner: Solutions Architect + Compliance Team

Security policies must be reviewed and approved by compliance and legal before Phase 4 activation begins.☐

Configure PCI-DSS 4.0 security zones and data classification policies.

REQUIRED

- Define cardholder data environment (CDE) scope in QBITEL Bridge policy engine
- Configure field-level sensitivity classification for ISO 8583 (PAN, CVV2, track data)
- Set encryption policy: ML-KEM-768 minimum for all CDE traffic
- Enable PCI-DSS compliance evidence collection (Requirements 3, 4, 8, 10)

☐

Configure DORA ICT resilience policies.

REQUIRED

- Define DORA major incident thresholds (transaction failure rate, latency SLA breach)
- Configure automatic DORA incident classification (critical, significant, major)
- Set DORA incident notification contacts (CISO, compliance, regulator email)
- Enable DORA Article 26 resilience testing schedule

☐

Configure SWIFT CSP security policies.

REQUIRED

- Set SWIFT environment boundary policy: only SWIFT Alliance Access IPs in SWIFT zone
- Configure SWIFT message signing policy: ML-DSA on all outbound MT/MX
- Enable SWIFT anomaly detection model with institution-specific baseline
- Set SWIFT CSP control evidence collection schedule

☐

Configure cryptographic agility policies.

REQUIRED

- Document all active cipher suites in QBITEL Bridge policy registry (PCI-DSS 12.3.3)
- Set algorithm deprecation schedule: RSA-2048 replacement timeline
- Enable automatic algorithm upgrade notification
- Configure quantum risk scoring threshold for mandatory escalation

☐

Configure autonomous incident response playbooks.

REQUIRED

- ISO 8583 velocity anomaly: auto-block + SOC alert
- SWIFT BIC deviation: auto-hold + CISO notification
- TN3270e unauthorized session: auto-terminate + audit log
- FIX sequence replay detected: auto-block + trading desk alert
- HSM tamper event: full system lockdown + emergency escalation



Conduct policy review with compliance team and legal.

REQUIRED

- Review autonomous response actions for legal and regulatory acceptability
- Confirm GDPR data minimization in logging configuration
- Confirm DORA incident classification thresholds with compliance officer

4

PHASE

PAYMENT RAIL PROTECTION

Weeks 6-10 | Owner: Solutions Architect + Head of Payments + Network Team

All payment rail activation steps require a pre-approved change request and out-of-hours maintenance window.☐

Activate ISO 8583 field-level encryption (scheduled maintenance window).

CRITICAL

- Pre-activation test: verify ISO 8583 proxy in pass-through mode with zero errors
- Cutover window: activate ML-KEM field encryption during off-peak (Sunday 02:00-04:00)
- Post-activation validation: end-to-end ISO 8583 test transaction through all schemes
- Monitor for 24 hours before confirming production stability

☐

Validate Visa and Mastercard scheme format compliance post-activation.

REQUIRED

- Run Visa Base I test suite against QBITEL Bridge proxy
- Run Mastercard Global Clearing test suite
- Confirm ISO 8583 MTI and bit-map structure unchanged
- Confirm PIN block encryption compatibility with payment HSM

☐

Activate SWIFT MT/MX PQC message signing.

CRITICAL

- Pre-activation: test ML-DSA signing on SWIFT test message with Alliance Access
- Coordinate with SWIFT Service Bureau for signing capability notification
- Activate ML-DSA signing on outbound SWIFT MT103, MT202, MT515
- Activate ML-KEM key exchange for SWIFT Alliance Access connectivity
- Monitor SWIFT NAK rates for 24 hours post-activation

☐

Activate FedWire and FedNow protection.

REQUIRED

- Notify Federal Reserve operational contact of PQC capability activation
- Activate PQC-enhanced TLS for Fedline Advantage connections
- Activate ML-DSA signing for FedNow ISO 20022 pacs.008 messages
- Validate FedNow 10-second processing SLA is maintained

☐

Activate ACH/NACHA file-level encryption.

REQUIRED

- Configure SFTP/MFT integration for PQC-protected NACHA file transmission
- Activate AES-256-GCM encryption with ML-KEM key exchange for ACH files
- Validate NACHA format integrity post-encryption with ACH operator
- Test both same-day ACH and standard ACH batch processing

☐ Activate CHIPS PQC authentication if applicable.

RECOMMENDED

- Coordinate with The Clearing House for PQC capability activation
- Activate ML-KEM key wrapping for CHIPS participant authentication

☐ Conduct 10,000 TPS performance validation under load.

REQUIRED

- Run 60-minute sustained load test at 10,000 TPS with ML-KEM active
- Measure p50, p95, p99 latency distribution - target p99 <50ms
- Measure HSM operation rate - target <20,000 ops/sec (within Luna HSM 7 spec)
- Confirm zero transaction errors during load test
- Document performance results for PCI-DSS evidence

☐ Train fraud analytics model on 90-day historical transaction data.

RECOMMENDED

- Load 90-day historical ISO 8583 transaction sample (anonymized)
- Train fraud detection model with institution-specific baseline
- Set fraud alert thresholds with Head of Fraud Risk
- Enable real-time fraud analytics in advisory mode (no auto-block) for first 30 days

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PHASE

MAINFRAME & LEGACY SHIELD

Weeks 8-12 | Owner: Solutions Architect + Mainframe Team

Mainframe deployments are zero code change. All interception occurs at network and gateway layers outside z/OS.

☐

Deploy TN3270e PQC proxy for branch teller network.

REQUIRED

- Configure QBITEL Bridge TN3270e proxy between teller LAN and CICS region
- Test with 10 pilot teller workstations before full rollout
- Verify CICS transaction codes function correctly through PQC proxy
- Verify 3270 screen rendering is unchanged for teller operators
- Rollout to all branch teller workstations in regional batches

☐

Deploy CICS transaction signing for external-facing transactions.

REQUIRED

- Identify CICS transactions handling external payment data
- Configure QBITEL Bridge CICS gateway interception for target transactions
- Test COMMAREA data integrity through signing layer
- Monitor CICS response times for <2ms overhead compliance

☐

Activate DB2 DRDA PQC upgrade.

REQUIRED

- Configure QBITEL Bridge DRDA proxy between distributed application servers and DB2
- Test DB2 application connectivity through DRDA proxy (JDBC test suite)
- Verify query response times are within <8ms overhead budget
- Monitor DB2 connection pool behavior through proxy

☐

Activate IBM MQ message signing.

REQUIRED

- Configure QBITEL Bridge MQ interception for queues carrying payment data
- Define MQ signing policy: which queue managers and channels require ML-DSA
- Test MQ message put/get through signing layer
- Verify MQ message format and MQMD headers are preserved

☐

Deploy AS/400 (IBM i) protection if applicable.

RECOMMENDED

- Configure TN5250e PQC proxy for IBM i terminal sessions
- Configure QBITEL Bridge for IBM i socket and JDBC connections
- Test RPG application connectivity through PQC proxy

☐ Validate mainframe performance: MIPS and response time.

REQUIRED

- Confirm MIPS consumption increase is <0.1% (zero direct mainframe impact)
- Confirm TN3270e session setup latency increase is <5ms
- Confirm CICS transaction response time increase is <2ms
- Confirm DB2 query response time increase is <8ms
- Document performance validation for compliance evidence

☐ VSAM file encryption (scheduled maintenance window).

RECOMMENDED

- Identify VSAM files containing sensitive payment data
- Schedule VSAM encryption activation during batch window
- Test COBOL program VSAM I/O through encryption layer
- Verify JCL and utility access to VSAM files is maintained

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PHASE

INTEGRATION DEPLOYMENT

Weeks 6-12 (parallel) | Owner: Solutions Architect + Integration Team

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Complete core banking system integration (Temenos/Finastra/FIS/Fiserv/Oracle).

REQUIRED

- Configure REST API integration for core banking event feeds
- Test payment message flow from core banking through QBITEL Bridge to payment gateway
- Verify account balance inquiry flows are protected
- Validate core banking vendor test certification (if required by contract)

☐

Complete HSM integration validation.

REQUIRED

- Run PKCS#11 compliance test suite against HSM cluster
- Verify ML-KEM key generation within HSM boundary
- Verify ML-DSA signing operations with HSM-stored keys
- Test HSM failover: pull primary HSM network cable, verify secondary takes over
- Verify key rotation completes without payment transaction interruption

☐

Complete SIEM/SOAR integration and playbook validation.

REQUIRED

- Validate all QBITEL Bridge alert types are received in SIEM
- Run playbook tests: inject test ISO 8583 anomaly, verify auto-response
- Run playbook tests: inject test SWIFT anomaly, verify alert chain
- Confirm ticket creation in ServiceNow/JIRA for all P0/P1 alerts
- Confirm DORA incident classification is correct for each alert type

☐

Complete fraud analytics integration with fraud operations team.

RECOMMENDED

- Deliver fraud analytics dashboard training to Head of Fraud Risk
- Configure fraud alert routing to fraud operations email/SMS
- Set dispute management system integration for auto-flagged transactions
- Define escalation path for fraud ML model threshold tuning

☐

Complete identity and access management (IAM) integration.

REQUIRED

- Configure CyberArk/BeyondTrust for QBITEL Bridge privileged access
- Integrate QBITEL Bridge management with corporate SSO (Okta/Azure AD)
- Configure QBITEL Bridge RBAC: Admin, Operator, Auditor, Read-Only roles
- Enable MFA requirement for all QBITEL Bridge management access



Complete compliance platform integration.

REQUIRED

- Configure GRC platform (RSA Archer/ServiceNow GRC/MetricStream) data feeds
- Validate PCI-DSS evidence package population in GRC
- Validate DORA ICT risk register population
- Configure compliance report scheduling (daily, weekly, monthly)

7
PHASE**MONITORING & ALERTING**

Weeks 10-13 | Owner: Solutions Architect + SOC Team

Monitoring must be operational for minimum 2 weeks before Phase 8 compliance validation begins.☐

Configure real-time transaction anomaly monitoring.

REQUIRED

- Set ISO 8583 velocity thresholds per card BIN and merchant category
- Set SWIFT MT103 amount deviation alerts (>3 standard deviations from 90-day baseline)
- Set TN3270e concurrent session anomaly alerts (>N sessions per operator ID)
- Set FIX sequence gap and replay alerts

☐

Configure PCI-DSS compliance monitoring dashboards.

REQUIRED

- Enable real-time PCI-DSS Requirement 3, 4, 8, 10 control status dashboard
- Configure encryption coverage report: % of CDE traffic with active PQC
- Set alert for any detected cleartext cardholder data in monitored segments
- Configure daily PCI-DSS compliance summary email to CISO

☐

Configure DORA incident reporting automation.

REQUIRED

- Enable DORA major incident auto-detection based on agreed thresholds
- Configure automated DORA incident report draft generation
- Set 4-hour notification SLA for DORA major incident classification
- Test DORA incident end-to-end from detection to report generation

☐

Configure HSM health monitoring.

REQUIRED

- Enable HSM capacity alerts (>80% operation utilization)
- Enable key expiration alerts (30-day advance warning)
- Enable HSM tamper event escalation to CISO and HSM vendor
- Configure HSM firmware version monitoring

☐

Configure cryptographic agility monitoring.

REQUIRED

- Enable alerts for detection of deprecated cipher suites (DES, 3DES, RC4, MD5)
- Enable alerts for RSA-1024 key detection in any protocol stream
- Monitor algorithm usage distribution report: % traffic on PQC vs classical



Conduct NOC team tabletop exercise.

REQUIRED

- Scenario 1: ISO 8583 payment rail attack - verify detection and response
- Scenario 2: SWIFT anomaly (potential fraud) - verify detection and escalation
- Scenario 3: HSM hardware failure - verify HA failover and payment continuity
- Scenario 4: DORA major incident - verify report generation and regulator notification

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PHASE

COMPLIANCE VALIDATION

Weeks 12-14 | Owner: Solutions Architect + Compliance Team + QSA

☐ PCI-DSS 4.0 evidence package preparation.

REQUIRED

- Requirement 3.5: Key management evidence (HSM custody, rotation logs)
- Requirement 4.2: Strong cryptography evidence (ML-KEM/ML-DSA deployment)
- Requirement 6.4: Web application protection evidence
- Requirement 8.6: MFA management evidence
- Requirement 10.2: Audit log evidence (tamper-evident, signed)
- Requirement 12.3.3: Cryptographic inventory and agility roadmap document

☐ Provide QSA with read-only dashboard access for evidence review.

REQUIRED

- Create QSA-role user account with read-only compliance portal access
- Provide QSA with 90-day historical compliance dashboard data
- Schedule QSA walkthrough of QBITEL Bridge architecture (2 hours)

☐ DORA ICT resilience test execution (Article 26).

REQUIRED

- Execute threat-led penetration test of ISO 8583 payment rail (QBITEL red team)
- Execute SWIFT message manipulation simulation
- Execute mainframe TN3270e session interception simulation
- Document resilience test results in DORA format
- Deliver DORA Article 26 test report to compliance officer

☐ SWIFT CSP attestation package preparation.

REQUIRED

- Generate SWIFT CSP control evidence report from QBITEL Bridge compliance portal
- Map QBITEL Bridge controls to SWIFT CSP 2025 mandatory and advisory controls
- Prepare CSP self-attestation supporting documentation

☐ DORA ICT risk register validation.

REQUIRED

- Review auto-populated DORA ICT risk register for completeness
- Validate ICT third-party risk entries for QBITEL Bridge and other ICT providers
- Obtain compliance officer sign-off on DORA risk register

☐

Basel III operational risk data validation.

RECOMMENDED

- Verify AMA loss data collection is operational
- Validate KRI feeds to operational risk platform
- Confirm BCBS 239 data lineage documentation for payment data flows

☐

Internal audit review.

REQUIRED

- Schedule internal audit review of QBITEL Bridge deployment
- Provide internal audit with SOC 2 Type II report and FIPS 140-3 certificate
- Facilitate internal audit access to compliance evidence portal

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PHASE

UAT & PERFORMANCE TESTING

Weeks 14-15 | Owner: Solutions Architect + QA + Payment Operations

☐

Execute full 10,000 TPS sustained load test.

REQUIRED

- Run 60-minute sustained load at 10,000 TPS with all PQC protections active
- Measure p50/p95/p99 latency at peak load - target p99 <50ms
- Measure error rate - target 0 transaction errors
- Measure HSM operation utilization - target <80% capacity
- Document results and sign off with Head of Payments

☐

Validate <50ms latency SLA across all protocol streams.

REQUIRED

- ISO 8583 field-level encryption: measure end-to-end latency for 1,000 sample transactions
- SWIFT MT103 with ML-DSA signing: measure added latency
- TN3270e session setup: measure time-to-screen for 100 sessions
- DB2 DRDA query: measure query response time delta vs baseline
- FIX order message signing: measure microsecond-level latency impact

☐

Execute HA failover test.

REQUIRED

- With active payment traffic flowing, physically disconnect primary QBITEL Bridge node
- Verify failover completes in <30 seconds
- Verify zero ISO 8583 transaction errors during failover
- Verify SWIFT connectivity maintained during failover
- Restore primary node and verify re-synchronization

☐

Execute HSM failover test.

REQUIRED

- With payment traffic flowing, disconnect primary HSM
- Verify QBITEL Bridge fails over to secondary HSM automatically
- Verify payment processing continues without interruption
- Verify HSM key material is consistent on secondary

☐

Execute bypass mode test.

REQUIRED

- Simulate QBITEL Bridge software failure (kill primary process)
- Verify bypass mode activates within configured timeout
- Verify payment traffic continues in bypass mode (unencrypted pass-through)
- Verify alert generated for bypass mode activation
- Verify QBITEL Bridge resumes active mode after process restart

☐

Execute fraud analytics validation.

RECOMMENDED

- Inject 100 synthetic fraudulent ISO 8583 transactions
- Verify fraud detection model flags >90% of injected fraud
- Verify false positive rate <0.5% on legitimate transaction baseline
- Validate fraud alert delivery to fraud operations team

☐

User acceptance sign-off from all stakeholders.

REQUIRED

- Head of Payments: payment rail protection and performance sign-off
- CISO: security controls and compliance evidence sign-off
- Head of Compliance: PCI-DSS and DORA evidence package sign-off
- Head of Technology: infrastructure and SLA sign-off
- Internal Audit: control evidence sign-off

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PHASE

GO-LIVE & HANDOVER

Week 16 | Owner: QBITEL NOC + Client Operations Team

Go-live requires signed UAT approval from all Phase 9 stakeholders before proceeding.☐

Execute go-live change request and CAB approval.

REQUIRED

- Submit formal change request for production go-live
- Obtain Change Advisory Board (CAB) approval
- Notify all stakeholders of go-live date and maintenance window
- Prepare rollback procedure document

☐

Activate all remaining protocol protections in sequence.

REQUIRED

- Final activation checklist walkthrough with QBITEL engineer
- Confirm all payment rails are in active PQC mode (not pass-through)
- Confirm all SWIFT connections are in signed mode
- Confirm all TN3270e sessions are in encrypted mode

☐

Deliver operations runbook to client NOC team.

REQUIRED

- QBITEL Bridge startup/shutdown procedures
- HA failover procedures (manual and automatic)
- HSM key rotation procedures
- Incident response playbooks (P0 through P3)
- DORA incident classification and reporting procedures
- PCI-DSS evidence generation procedures
- Vendor escalation contacts and SLA reference

☐

Establish 24/7 NOC support handover.

REQUIRED

- QBITEL NOC contact numbers and escalation path
- Client NOC primary and secondary contacts registered with QBITEL
- PagerDuty/OpsGenie integration for P0 alert escalation
- Regular NOC sync schedule (weekly for first 3 months)

☐

Deliver training to all operational teams.

REQUIRED

- QBITEL Bridge operator certification: 1-day final assessment
- SOC analyst refresher: 4-hour session on production alert handling
- Compliance team: 2-hour session on evidence portal and report generation
- Executive dashboard: 1-hour session for CISO/CTO on management reporting

☐

Complete project sign-off and handover documentation.

REQUIRED

- Project completion certificate signed by client project owner
- As-built architecture diagram delivered and verified
- All change requests closed in client ITSM system
- Warranty period start date confirmed (default: 90 days)

☐

Schedule 30-day and 90-day post-go-live health checks.

REQUIRED

- 30-day check: performance review, compliance evidence review, alert tuning
- 90-day check: fraud model performance review, compliance evidence package review

Appendix A: Performance SLAs

Metric	SLA Target	Measurement Method	Breach Consequence
System availability	99.999% (5 nines)	QBITEL NOC uptime monitoring	Credit: 10x monthly subscription
ISO 8583 throughput	10,000 TPS minimum	Monthly load test	Credit: 5x daily subscription
ISO 8583 p99 latency	<50ms end-to-end	Continuous p99 measurement	Credit: 2x daily subscription
SWIFT added latency	<8ms per message	SWIFT message timing logs	Credit: 2x daily subscription
FIX added latency	<50 microseconds	FIX session timing	Credit: 2x daily subscription
TN3270e session setup	<5ms added	Session timing logs	Notification only
HA failover time	<30 seconds	Quarterly failover test	Credit: 5x daily subscription
P0 incident response	<5 min engineer engaged	NOC escalation logs	Credit: 10x daily subscription
Key rotation	Zero downtime	Rotation event logs	Credit: 5x daily subscription

Appendix B: Rollback Procedures

Scenario	Rollback Action	Time to Rollback	Payment Impact
ISO 8583 PQC failure	Set payment-rail policy to PASSTHROUGH mode	<30 seconds	Zero - pass-through maintains connectivity
SWIFT signing failure	Disable ML-DSA signing policy for SWIFT zone	<30 seconds	SWIFT messages continue unsigned
TN3270e proxy failure	Set TN3270e policy to DIRECT CONNECT mode	<60 seconds	Teller sessions reconnect directly to CICS
Full QBITEL Bridge failure	Remove inline deployment; bypass with patch cable	15 minutes	Full bypass - no PQC protection
HSM failure (primary)	Automatic failover to secondary HSM	<10 seconds	Zero - automatic
HSM failure (both)	Activate emergency key cache from encrypted backup	2 minutes	Degraded - local key cache only

Appendix C: Escalation Matrix

Incident Priority	Definition	Response Time	QBITEL Escalation	Client Escalation
P0 - Critical	Active payment rail attack or complete system failure	<5 minutes engineer	NOC Lead + VP Engineering	CISO + CTO + Head of Payments
P1 - High	SWIFT anomaly, significant performance degradation	<15 minutes engineer	SSC alarm NOC Engineer	CISO + Head of Payments
P2 - Medium	Compliance violation, non-critical alerts, key outage	<1 hour response	NOC Engineer	Head of Compliance + IT Manager
P3 - Low	Informational alerts, performance optimization, feature requests	Next business day	Support Engineer	IT Operations Manager
Regulatory	DORA major incident, PCI-DSS breach, information breach	Immediate parallel	QBITEL Compliance + NOC	CISO + CCO + Legal + Regulator